

Assessment Growth and Section Performance Analytics in R

Public-Safe BOY/EOY Improvement Study

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Recommendation

Use the latest completed assessment year, **2024-2025**, to identify teacher, course, and section patterns that deserve review before the next cycle. The stakeholder metric is **BOY/EOY score gain**: end-of-year score minus beginning-of-year score for the same public-safe student record.

The decision system does not ask whether old sections should be managed retroactively. It asks whether the most recent actual growth was materially above or below an expected-growth baseline trained only on prior years.

How To Read The Model Results

The model is **not** intended to forecast each student's exact score gain. Individual growth is noisy because similar students can improve by different amounts for reasons not fully captured in the extract. For that reason, the latest-year individual gain R-squared of **0.097** is treated as a signal-strength diagnostic, not the headline business result.

The business question is group-level: after adjusting for starting score, readiness, attendance, prior history, course track, grade level, and section composition, did a teacher, course, or section produce more or less average growth than expected? That comparison is more stable because student-level noise partly averages out across groups.

The EOY R-squared of **0.932** is reported only to show that final score is mechanically easier to predict from BOY score. It should not be read as evidence that the model precisely predicts improvement. The relevant evidence is out-of-sample lift versus a naive baseline, aggregate fit, residual calibration, uncertainty intervals, and flag stability.

The baseline selected for operations is **Growth lasso**. It predicts **score gain directly** using a Regularized regression specification. The selected tuning choice is $\alpha=1.00$. Against a naive prior-year mean-growth baseline, the selected model improves temporal RMSE by **8.0%** and latest-year RMSE by **5.4%**.

Under the stricter validity framework, this model is best described as **directional review** evidence: 4 of 7 primary gates passed. The model is useful for structured review and prioritization, but the report should not overstate it as a definitive teacher or section rating system.

Because decisions are made at aggregate review levels, the planning-level fit is more relevant than the individual gain R-squared: latest-year gain R-squared is **0.266** for section means, **0.546** for course means, and **0.638** for teacher means.

The workflow uses the direct-growth model to create a fair expected-growth baseline, then makes decisions from aggregate teacher, course, and section residuals with bootstrap uncertainty checks.

Slice	Priority or watch rows	Total reviewed
Teachers	1	5
Courses	3	9
Sections	8	24

Decision	Slice	Target	N	Gap	95% CI	q
Intervention	Section	S21	12	-2.67	-5.15 to -0.31	0.200
Intervention	Course	Precalc	40	-1.41	-2.88 to -0.25	0.040

Bright spots, watch-list rows, and the full decision table are generated as [decision table](#).

Plain-English Method

1. Build a paired BOY/EOY growth extract from public-safe assessment records.
2. Define the business outcome as score gain: EOY score minus BOY score.
3. Create leakage-safe engineered features from BOY data and prior completed years: multi-year student history, section composition, transformations, and interaction terms.
4. Search candidate expected-growth models across parametric, spline, GAM, regularized, tree-based, forest, boosting, MARS, ensemble, and excluded leakage-check specifications.
5. Select the operating baseline using rolling-origin temporal validation, with repeated CV, process validation, locked-holdout review, bootstrap checks, and feature-stability diagnostics as supporting evidence.
6. Score the latest year against the prior-year baseline.
7. Aggregate observed-minus-expected growth by teacher, course, and section.
8. Flag review targets only when the gap is large enough to matter and the uncertainty check supports follow-up.

This design separates the prediction problem from the decision problem. The prediction model estimates what growth would be expected for a similar starting profile; the review layer asks where actual growth departed from that expectation.

Direct Answers

1. The analysis covers 1,737 paired BOY/EOY records across 174 section-year groups and 5 public-safe teacher identifiers.
2. The training window is 2018-2019, 2019-2020, 2020-2021, 2021-2022, 2022-2023, 2023-2024; the action year is 2024-2025.
3. The average raw gain across the full extract is 5.72 points; the latest-year raw gain is 5.34 points.
4. The model search tested 31 candidate baselines across parametric, nonlinear, ensemble, and leakage-check families.
5. The selected direct-growth baseline has temporal expected-gain RMSE 4.663, temporal MAE 3.725, latest-year RMSE 4.693, and latest-year MAE 3.739.

6. The modest individual gain R-squared is expected for a noisy improvement outcome; it is not the main business score.
7. Teacher, course, and section flags are audit priorities. They are not automatic personnel ratings or causal claims.

Data Audit

A record enters the growth model only when the same public-safe student has valid BOY and EOY scores in the same section and teacher context. This keeps improvement tied to one instructional experience instead of mixing students across sections.

Measure	Value
Raw assessment rows	4,018
BOY/EOY candidate pairs	2,009
Included paired records	1,737
Unique public-safe student IDs	671
Unique section-year groups	174
Unique simulated teachers	5
Records with prior-year history	1,066
Mean section size	10.4
Mean BOY score	48.5
Mean EOY score	54.2
Mean BOY/EOY gain	5.7
Median section paired records	10

Model Selection

The model discovery system used 1,485 prior-year pairs and held out 252 latest-year pairs for action-year evaluation. The primary selection metric is rolling-origin temporal expected-gain RMSE, not latest-year performance, so each validation year is treated like the future.

The best eligible direct-growth rolling-origin RMSE was **4.663** from **Growth lasso**, so it is the operating baseline. Repeated-CV RMSE remains a stability check for candidates that are practically tied on rolling-origin RMSE.

The model-search guardrails were:

- Use direct BOY/EOY score gain as the operating target because that is the stakeholder performance metric.
- Select by rolling-origin temporal RMSE so the baseline is judged on future-facing generalization.
- Use repeated-CV RMSE as the tie-breaker when rolling-origin RMSE differs by less than 0.01 points.
- Keep teacher, course, and section identifiers out of the operating baseline because those are the review slices.
- Use feature engineering only when the feature is available at BOY or from prior completed years.
- Refit the selected production model on all completed years only after model selection.
- Report individual gain R-squared as a signal-strength diagnostic, not as the headline business result.
- Report EOY R-squared only as context because final score is mechanically easier to predict than growth.

The decision-grade target table below is intentionally strict. It prevents the project from claiming stronger performance judgment than the data support.

Gate	Decision-grade	Actual	Status
Rolling RMSE lift vs naive	10.0%	8.0%	Below decision-grade
Rolling MAE lift vs naive	8.0%	8.6%	Pass
Temporal gain R-squared	0.200	0.151	Below decision-grade
Section mean gain R-squared	0.300	0.266	Below decision-grade
Course mean gain R-squared	0.500	0.546	Pass
Teacher mean gain R-squared	0.600	0.638	Pass
Overall residual bias	0.200	-0.258	Review
Maximum subgroup residual bias	0.500	3.252	Review

The first strength check is whether the selected model beats a naive baseline that predicts the training-year mean gain for every record. This is the minimum bar for using a model as an expected-growth baseline.

Measure	Value
Naive temporal RMSE	5.066
Selected temporal RMSE	4.663
Temporal RMSE improvement	8.0%
Naive latest-year RMSE	4.960
Selected latest-year RMSE	4.693
Latest-year RMSE improvement	5.4%
Naive latest-year MAE	4.026
Selected latest-year MAE	3.739
Latest-year MAE improvement	7.1%
Latest-year gain R-squared	0.097
Latest-year EOY R-squared	0.932
Section-mean gain R-squared	0.266
Teacher-mean gain R-squared	0.638
Course-mean gain R-squared	0.546

The table below shows the strongest candidate baselines tested. The selected model was chosen by rolling-origin validation, not by whichever model looked best on the latest year. The leakage-check row is shown for transparency but is not eligible because it would absorb the teacher/course patterns the review layer is designed to detect.

Candidate model	Model type	Used?	Rolling RMSE	Latest RMSE	Latest MAE	Gain R2	Read
Growth lasso	Regularized	Yes	4.663	4.693	3.739	0.097	Selected
Growth elastic net	Regularized		4.666	4.699	3.740	0.095	Near tie
Growth linear	Linear		4.698	4.683	3.728	0.101	Near tie
GBM 1	Boosting		4.702	4.727	3.814	0.084	Near tie
GBM 2	Boosting		4.709	4.695	3.780	0.097	Near tie
MARS 2	MARS		4.725	4.618	3.714	0.126	Near tie
Ranger 1	Ranger		4.740	4.685	3.732	0.100	Near tie
Leakage check	Leakage		2436.990	4.658	3.708	0.111	Leakage only

Full model artifacts: [model comparison](#), [model search grid](#), [model strength](#), [family summary](#), [selection rationale](#), [temporal validation](#), [rolling-origin validation](#), [process validation](#), [locked holdout](#), [validity targets](#), and [bootstrap validation](#).

Feature Discovery

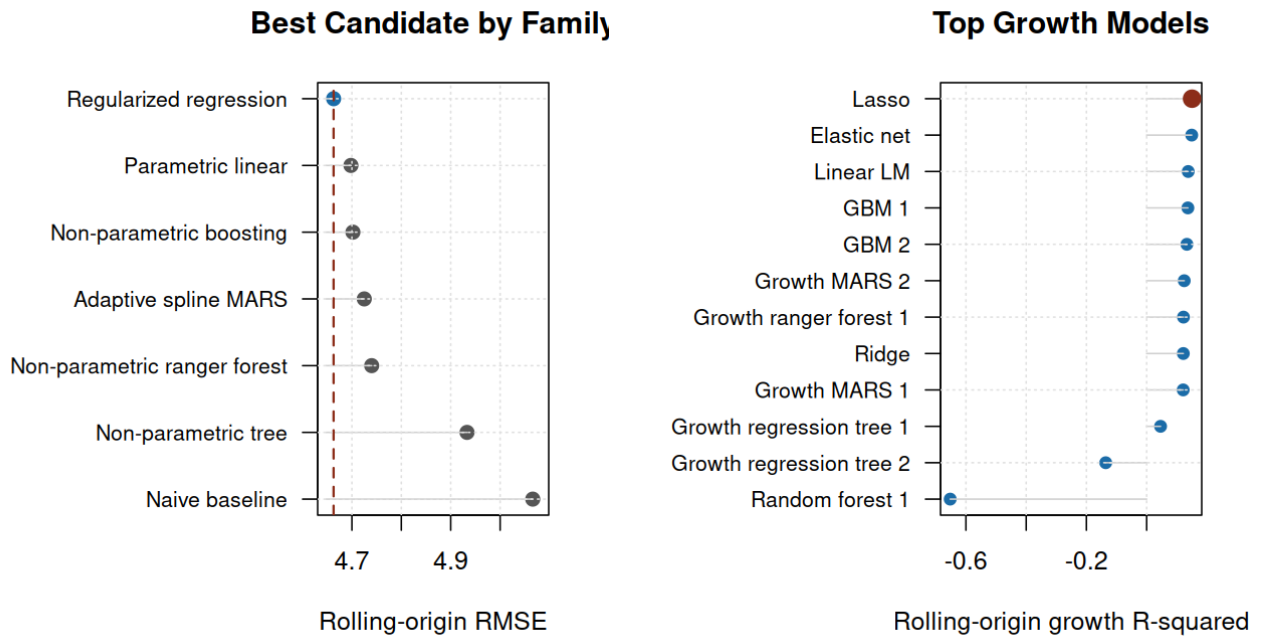
The model search includes leakage-safe feature engineering: multi-year prior student growth, prior trend and volatility, BOY score bands, section composition, attendance mix, nonlinear basis terms, and selected interactions. The table below shows which features mattered most when permuted in the locked action-year data.

Feature	RMSE lift if permuted	SD
course track	0.056	0.034
student prior year count	0.022	0.009
grade level	0.015	0.006
section pct high absence	0.006	0.002
student prior attendance mean	0.000	0.000
boy readiness	0.000	0.000
boy score	0.000	0.000
school year offset	0.000	0.000
section attendance mean	0.000	0.000
section boy mean	0.000	0.000

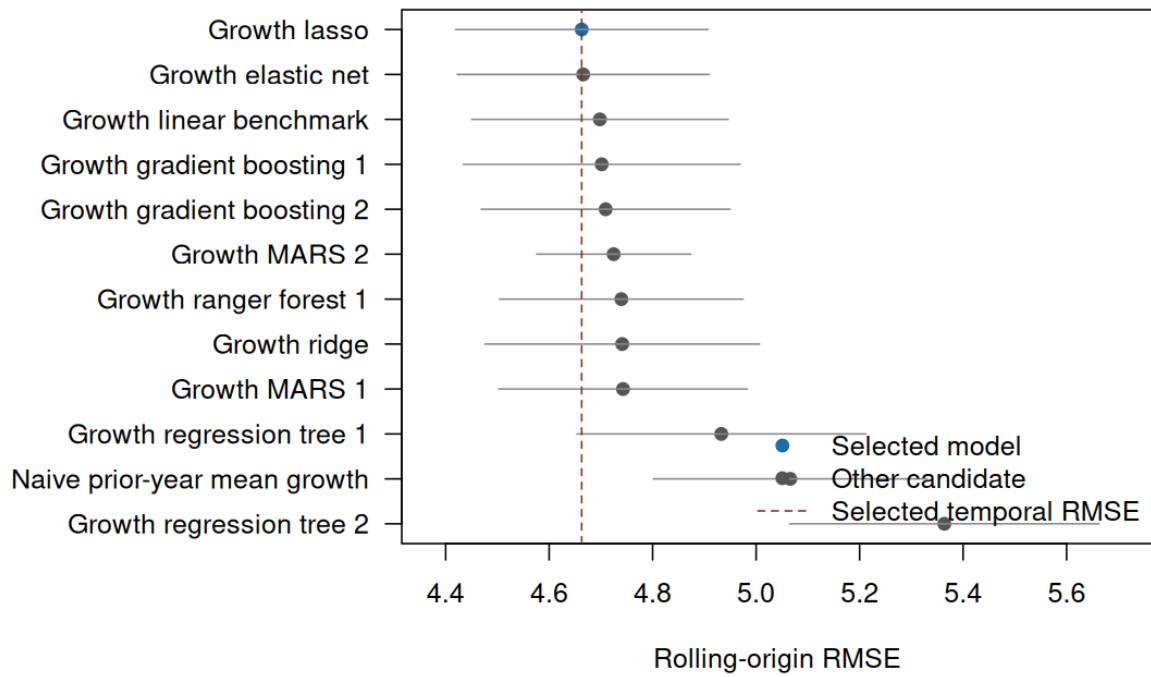
Feature stability asks whether the same predictors continue to matter across repeated perturbations.

Feature	Positive importance	Top-quartile importance
course track	88%	88%
student prior year count	100%	100%
grade level	100%	100%
section pct high absence	100%	100%
student prior attendance mean	62%	62%
boy readiness	0%	100%
boy score	0%	100%
school year offset	0%	100%
section attendance mean	0%	100%
section boy mean	0%	100%

Feature artifacts: [feature importance](#) and [feature stability](#).



Expected-Growth Model Comparison



Latest-Year Review Targets

The latest-year review layer compares observed gain with expected gain for 24 section groups. The decision labels use a practical audit threshold: material gap, bootstrap interval direction, and BH-adjusted q-value for multiple-review control.

Teacher review

Target	N	Raw	Expected	Gap	CI	q	Decision
TCH-005	68	3.81	4.80	-0.99	-2.19 to 0.06	0.367	Watch

Course review

Target	N	Raw	Expected	Gap	CI	q	Decision
Precalc	40	4.99	6.41	-1.41	-2.88 to -0.25	0.040	Intervention
AP Precalc	30	6.11	4.18	+1.93	0.30 to 3.20	0.030	Bright spot
AP Calc AB	49	4.31	5.17	-0.86	-2.23 to 0.37	0.384	Watch

Section evidence

Target	N	Raw	Expected	Gap	CI	q	Decision
S21	12	3.10	5.77	-2.67	-5.15 to -0.31	0.200	Intervention
S02	10	9.02	6.56	+2.46	0.16 to 4.92	0.200	Bright spot
S13	9	7.18	4.28	+2.90	0.98 to 4.80	0.040	Bright spot
S19	12	2.41	5.01	-2.60	-5.04 to 0.12	0.320	Watch
S04	9	4.01	6.45	-2.44	-6.44 to 2.33	0.556	Watch
S16	10	4.11	6.27	-2.15	-5.05 to 0.27	0.347	Watch
S17	9	4.78	6.72	-1.94	-5.60 to 1.70	0.556	Watch
S23	9	1.89	3.75	-1.87	-4.37 to 0.64	0.366	Watch

Full review tables: [teacher review](#), [course review](#), and [section review](#).

A second review layer fits a mixed-effects shrinkage model on the latest-year residuals. It estimates teacher, course, and section effects at the same time and pulls noisier small-group estimates toward zero, so the strongest flags are less likely to be one-off small-section artifacts.

Slice	Target	N	Raw gap	Shrunk gap	95% CI	q	Decision
Course	AP	30	+1.93	+0.66	-0.36 to 1.69	0.869	In range
Course	Precalc	40	-1.41	-0.43	-1.40 to 0.54	0.869	In range
Section	S02	10	+2.46	+0.34	-0.68 to 1.36	0.965	In range
Section	S19	12	-2.60	-0.31	-1.32 to 0.70	0.965	In range
Section	S21	12	-2.67	-0.30	-1.31 to 0.70	0.965	In range
Course	AP Calc AB	49	-0.86	-0.30	-1.24 to 0.63	0.869	In range
Section	S18	13	+1.33	+0.29	-0.72 to 1.29	0.965	In range
Section	S13	9	+2.90	+0.26	-0.76 to 1.28	0.965	In range
Section	S04	9	-2.44	-0.25	-1.27 to 0.78	0.965	In range
Section	S06	11	+1.13	+0.19	-0.82 to 1.20	0.965	In range
Course	Alg 2 H	26	+0.32	+0.18	-0.87 to 1.23	0.869	In range
Section	S23	9	-1.87	-0.17	-1.20 to 0.85	0.965	In range

Flag stability estimates how often a slice remains beyond the practical one-point gap threshold under bootstrap resampling. Directional rows should be reviewed with context; stable rows have stronger evidence for action.

Level	Target	N	Adjusted gap	Decision	Stability	Status
Section	Y06-SEC-013 / MATH-AP-PRECALC / TCH-004	9	+2.90	Positive anomaly	96%	Stable
Section	Y06-SEC-021 / MATH-AP-CALC-AB / TCH-005	12	-2.67	Intervention target	90%	Stable
Section	Y06-SEC-019 / MATH-AP-CALC-AB / TCH-005	12	-2.60	Watch list	89%	Stable

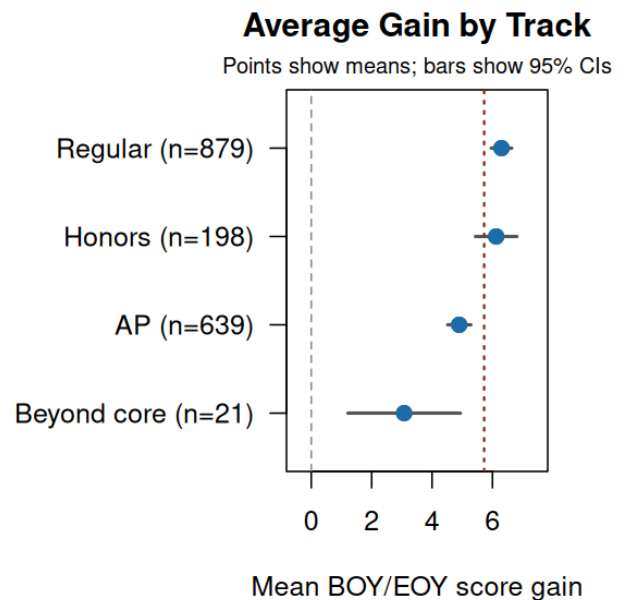
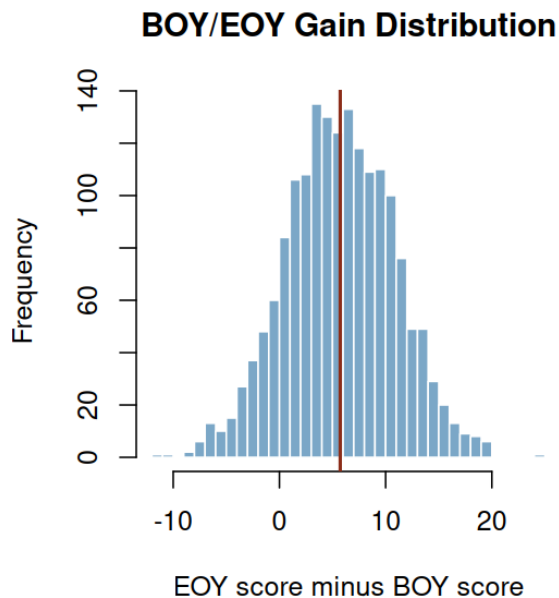
Level	Target	N	Adjusted gap	Decision	Stability	Status
Course	MATH-AP-PRECALC	30	+1.93	Positive anomaly	88%	Stable
Section	Y06-SEC-002 / MATH-GEOM / TCH-002	10	+2.46	Positive anomaly	88%	Stable
Section	Y06-SEC-016 / MATH-PRECALC / TCH-004	10	-2.15	Watch list	82%	Stable
Section	Y06-SEC-011 / MATH-AP-PRECALC / TCH-004	10	+2.07	Watch list	78%	Stable
Section	Y06-SEC-023 / MATH-AP-CALC-BC / TCH-005	9	-1.87	Watch list	77%	Stable
Section	Y06-SEC-004 / MATH-GEOM / TCH-002	9	-2.44	Watch list	74%	Stable
Course	MATH-PRECALC	40	-1.41	Intervention target	69%	Directional

Shrinkage artifacts: [shrinkage status](#) and [shrinkage review](#). Flag-stability artifacts: [flag stability](#).

Section Evidence

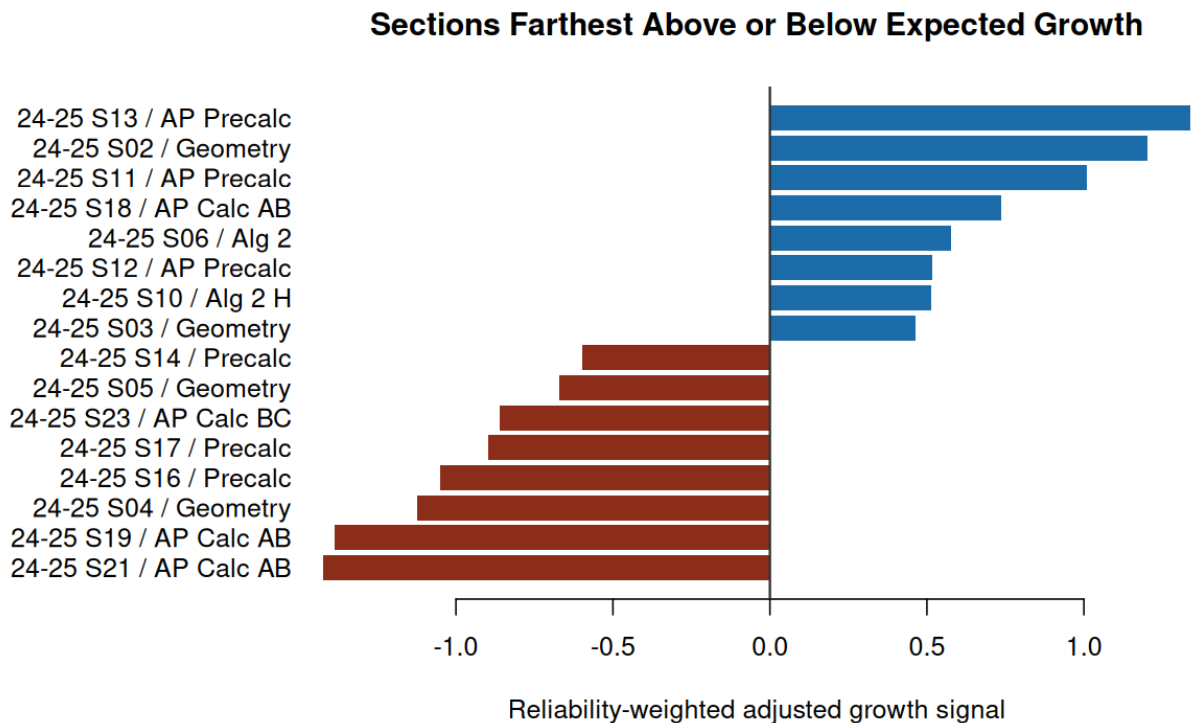
Raw section improvement is useful for communication, but it is not the final comparison. A section can show positive raw gain and still fall below expected growth if its starting profile suggested a larger increase.

Section	Course	N	BOY	EOY	Gain	95% CI	p-value
S13	AP Precalc	9	63.5	70.7	7.18	5.20 to 9.16	<0.001
S02	Geometry	10	40.1	49.1	9.02	6.32 to 11.73	<0.001
S11	AP Precalc	10	69.0	74.7	5.67	1.52 to 9.83	0.013
S18	AP Calc AB	13	49.4	56.1	6.69	3.06 to 10.31	0.002
S06	Alg 2	11	41.0	48.6	7.63	4.97 to 10.30	<0.001
S12	AP Precalc	11	59.5	65.1	5.64	1.86 to 9.43	0.008
S10	Alg 2 H	12	63.1	70.0	6.90	3.98 to 9.83	<0.001
S03	Geometry	9	43.2	50.2	7.02	3.12 to 10.92	0.003



The adjusted section signal is observed gain minus expected gain, reliability-weighted toward zero for smaller sections.

Section	Teacher	Course	N	Raw	Expected	Signal	Result
S13	TCH-004	AP Precalc	9	7.18	4.28	+1.34	Above
S02	TCH-002	Geometry	10	9.02	6.56	+1.20	In range
S11	TCH-004	AP Precalc	10	5.67	3.60	+1.01	In range
S18	TCH-005	AP Calc AB	13	6.69	5.35	+0.74	In range
S17	TCH-003	Precalc	9	4.78	6.72	-0.90	In range
S23	TCH-005	AP Calc BC	9	1.89	3.75	-0.86	In range
S05	TCH-001	Geometry	11	4.00	5.31	-0.67	In range
S14	TCH-003	Precalc	10	4.81	6.04	-0.60	In range



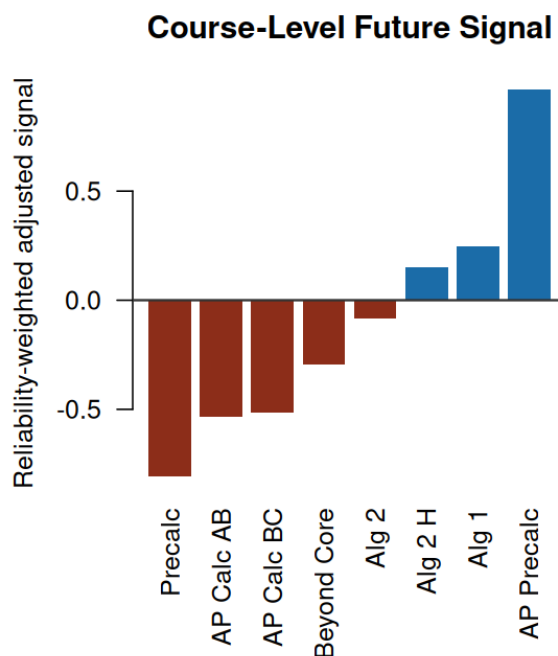
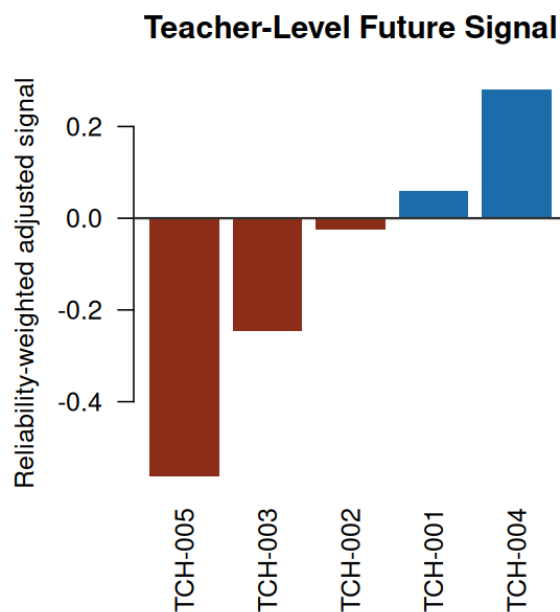
The full section signal table is generated as [section signals](#).

Teacher and Course Summaries

These summaries aggregate the latest-year evidence into planning views. They support review conversations about pacing, curriculum alignment, attendance mix, and transferable practices. They should not be read as standalone personnel scores.

Teacher	Sections	Records	Raw	Expected	Signal
TCH-004	5	43	5.21	4.59	+0.28
TCH-001	3	33	6.06	5.90	+0.06
TCH-002	5	52	6.60	6.65	-0.02
TCH-003	5	56	5.70	6.18	-0.25
TCH-005	6	68	3.81	4.80	-0.56

Course	Sections	Records	Raw	Expected	Signal
AP Precalc	3	30	6.11	4.18	+0.97
Alg 1	1	13	7.13	6.32	+0.24
Alg 2 H	2	26	6.18	5.86	+0.15
Geometry	4	39	5.99	6.06	-0.04
Alg 2	3	33	6.57	6.73	-0.08
Beyond Core	1	3	-0.11	3.14	-0.30
AP Calc BC	2	19	2.52	3.85	-0.51
AP Calc AB	4	49	4.31	5.17	-0.54
Precalc	4	40	4.99	6.41	-0.81



Diagnostics and Sensitivity

The baseline is strong for final-score expectation and weaker for individual gain variation. That pattern is expected and is part of the interpretation, not something to hide: BOY score explains much of EOY score, while individual improvement contains more unobserved classroom, attendance, motivation, and assessment noise. For review decisions, the model is used to build expected growth and then aggregate residuals by teacher, course, and section.

Diagnostic	Estimate	Interpretation
Latest-year expected-gain RMSE	4.693	Typical out-of-sample prediction error on latest-year BOY/EOY gain
Latest-year expected-gain R-squared	0.097	Share of latest-year gain variation explained by the expected-growth model
Latest-year EOY R-squared	0.932	Share of latest-year EOY score variation explained by the baseline
Latest-year residual mean	-0.258	Near 0 means expected gain is centered in the action year
Latest-year residual SD	4.695	Latest-year residual spread used for slice uncertainty

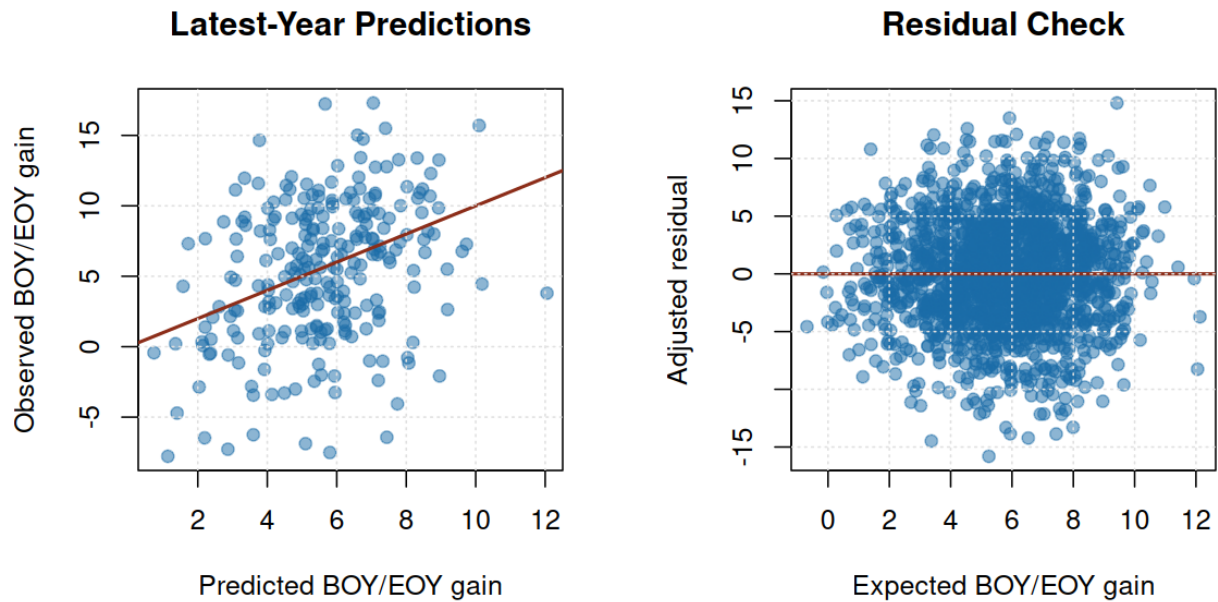
Metric	Estimate	95% interval
Expected-gain RMSE	4.693	4.305 to 5.085
Expected-gain MAE	3.739	3.377 to 4.079
Expected-gain R-squared	0.097	-0.005 to 0.181
EOY RMSE	4.693	4.305 to 5.085
EOY R-squared	0.932	0.917 to 0.943

Metric	Value
Selected locked-holdout RMSE	4.693
Naive locked-holdout RMSE	4.960
Locked-holdout RMSE lift	5.4%
Selected locked-holdout MAE	3.739
Naive locked-holdout MAE	4.026
Locked-holdout MAE lift	7.1%
Locked-holdout gain R-squared	0.097

Diagnostic	Value
Individual gain SD	5.079
Student mean-gain R-squared	0.208
Section mean-gain R-squared	0.266
Course mean-gain R-squared	0.546

Diagnostic	Value
Teacher mean-gain R-squared	0.638
Selected train-validation R-squared gap	0.045

Measure	Value
Training paired records	1,485
Latest-year paired records	252
Latest-year section-year groups	24
Latest-year mean raw BOY/EOY gain	5.34
Latest-year raw-vs-adjusted rank correlation	0.826
Latest-year top-10 overlap, raw vs adjusted ranking	70.0%



Technical Appendix

The operating model excludes teacher IDs, course IDs, and section IDs. A leakage benchmark is reported only to show what would happen if persistent IDs were included in the baseline; it is not used for review because it would absorb the teacher/course patterns the decision layer is designed to detect.

Package	Installed
mgcv	Available
randomForest	Available
ranger	Available
gbm	Available
rpart	Available
glmnet	Available
earth	Available

Package	Installed
lme4	Available

Metric	Value
Selected model	Growth lasso
Selected target strategy	Direct growth
Selected method	Elastic net
Selected family	Regularized regression
Selected tuned parameters	alpha=1.00
Selection rule	Lowest rolling-origin temporal RMSE among eligible direct-growth candidates; repeated-CV RMSE is the tie-breaker within 0.01 points
Training paired records	1,485
Latest-year action paired records	252
Training years	2018-2019, 2019-2020, 2020-2021, 2021-2022, 2022-2023, 2023-2024
Action year	2024-2025
Candidate models tested	31
Operational candidates tested	26
Excluded leakage benchmarks	5
Repeated CV folds	5
Repeated CV repeats	2
Temporal expected-gain RMSE	4.663
Temporal expected-gain MAE	3.725
Temporal expected-gain R-squared	0.151
Temporal expected-gain RMSE SD	0.244
Temporal EOY R-squared	0.934
Latest-year expected-gain RMSE	4.693
Latest-year expected-gain MAE	3.739
Latest-year expected-gain R-squared	0.097
Latest-year EOY R-squared	0.932

The full selected-model metrics table is generated as [selected-model metrics](#).

Conclusion

The project should be read as a statistical decision-support system, not as a simple prediction demo. The strongest business value is the workflow: choose a validated expected-growth baseline, compare latest actual growth to that baseline at the group level, quantify uncertainty by slice, and translate the evidence into review priorities.

The recommended stakeholder action is to review the flagged teacher, course, and section patterns before the next assessment cycle. Priority targets deserve support or investigation; positive anomalies deserve study for transferable practices; watch-list rows deserve context review before escalation.

The important limitation is that the data are public-safe and generalized from an assessment workflow. The outputs demonstrate the analysis pattern and should not be used as real student, teacher, or personnel decisions.

Reproducibility

Rebuild the full evidence packet with `make all`. The pipeline uses the included public-safe extract and no credentials, private files, or network access.

Public-Safety Statement

This report is an original public-safe portfolio artifact. It excludes private coursework prompts, exams, rubrics, syllabi, lecture transcripts, source datasets, personal data, real student-identifiable records, real personnel records, credentials, and copyrighted source documents.